

General Relativity HW-4

PROBLEM 1. Derive a variation of the Riemann tensor which should become

$$\delta R_{\alpha\nu\beta}^{\mu} = D_{\nu}\gamma_{\alpha\beta}^{\mu} - D_{\beta}\gamma_{\alpha\nu}^{\mu}$$

where $\gamma_{\alpha\beta}^{\mu} = \delta\Gamma_{\alpha\beta}^{\mu} = \frac{1}{2}(D_{\alpha}h_{\beta}^{\mu} + D_{\beta}h_{\alpha}^{\mu} - D^{\mu}h_{\alpha\beta})$ and D_{α} is a covariant derivative. Show that then we have

$$\delta R_{\alpha\beta} = \frac{1}{2}(D_{\nu}D_{\alpha}h_{\beta}^{\nu} + D_{\nu}D_{\beta}h_{\alpha}^{\nu} - \square h_{\alpha\beta} - D_{\alpha}\partial_{\beta}h)$$

and

$$\delta R = (-R_{\mu\nu} + D_{\mu}D_{\nu} - g_{\mu\nu}\square)h^{\mu\nu}$$

where $\square = g^{\mu\nu}D_{\mu}D_{\nu}$ is the d'Alembertian operator and $h = h^{\mu}_{\mu}$.

Extra Task for extra 10 points. Derive the second variation of the Einstein-Hilbert action around Minkowski space-time in 4 dimensions. The result should be

$$\delta^2 S = \int d^4x \sqrt{-\eta} \frac{M_P^2}{2} \left[\frac{1}{4} h_{\mu\nu} \square h^{\mu\nu} - \frac{1}{4} h \square h + \frac{1}{2} h \partial_{\mu} \partial_{\nu} h^{\mu\nu} + \frac{1}{2} \partial_{\mu} h^{\mu\rho} \partial_{\nu} h_{\rho}^{\nu} \right]$$

SOLUTION:

1.

$$g_{\mu\nu} \rightarrow g_{\mu\nu} + h_{\mu\nu}, g^{\mu\nu} \rightarrow g^{\mu\nu} - h^{\mu\nu}$$

Remember the definition of Riemann tensor:

$$R^{\mu}_{\alpha\nu\beta} V^{\alpha} = [D_{\nu}, D_{\beta}] V^{\mu}$$

When letting $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, we have:

$$\delta D_{\mu} V^{\beta} = \delta(\partial_{\mu} + \Gamma_{\mu\alpha}^{\beta}) V^{\alpha} = \delta \Gamma_{\mu\alpha}^{\beta} V^{\alpha}$$

Thus,

$$\begin{aligned} \delta R^{\mu}_{\alpha\nu\beta} V^{\alpha} &= \delta D_{\nu}(D_{\beta} V^{\mu}) + D_{\nu}(\delta D_{\beta} V^{\mu}) - \delta D_{\beta} D_{\nu} V^{\mu} - D_{\beta}(\delta D_{\nu} V^{\mu}) \\ &= \gamma_{\nu\alpha}^{\mu} D_{\beta} V^{\alpha} - \gamma_{\beta\nu}^{\alpha} D_{\alpha} V^{\mu} + D_{\nu}(\gamma_{\beta\alpha}^{\mu} V^{\alpha}) - (\gamma_{\beta\alpha}^{\mu} D_{\nu} V^{\alpha} - \gamma_{\nu\beta}^{\alpha} D_{\alpha} V^{\mu} + D_{\beta}(\gamma_{\nu\alpha}^{\mu} V^{\alpha})) \\ &= \gamma_{\nu\alpha}^{\mu} D_{\beta} V^{\alpha} - \gamma_{\beta\nu}^{\alpha} D_{\alpha} V^{\mu} + D_{\nu}(\gamma_{\beta\alpha}^{\mu} V^{\alpha}) + \gamma_{\beta\alpha}^{\mu} D_{\nu} V^{\alpha} - (\gamma_{\beta\alpha}^{\mu} D_{\nu} V^{\alpha} - \gamma_{\nu\beta}^{\alpha} D_{\alpha} V^{\mu} + D_{\beta}(\gamma_{\nu\alpha}^{\mu} V^{\alpha}) + \gamma_{\nu\alpha}^{\mu} D_{\beta} V^{\alpha}) \\ &= (D_{\nu}\gamma_{\alpha\beta}^{\mu} - D_{\beta}\gamma_{\alpha\nu}^{\mu}) V^{\alpha} \end{aligned}$$

where we have used the fact that $\gamma_{\alpha\beta}^{\mu}$ is symmetric with respect to α, β .

Thus

$$\delta R_{\alpha\nu\beta}^{\mu} = D_{\nu}\gamma_{\alpha\beta}^{\mu} - D_{\beta}\gamma_{\alpha\nu}^{\mu}$$

□

2.

$$\begin{aligned}\delta R_{\alpha\beta} &= \delta R^\nu_{\alpha\nu\beta} = D_\nu \gamma^\nu_{\alpha\beta} - D_\beta \gamma^\nu_{\alpha\nu} = \frac{1}{2}(D_\nu D_\alpha h^\nu_\beta + D_\nu D_\beta h^\nu_\alpha - D_\nu D^\nu h_{\alpha\beta}) - \frac{1}{2}D_\beta(D_\alpha h^\nu_\nu + D_\nu h^\nu_\alpha - D^\nu h_{\alpha\nu}) \\ &= \frac{1}{2}(D_\nu D_\alpha h^\nu_\beta + D_\nu D_\beta h^\nu_\alpha - \square h_{\alpha\beta}) - \frac{1}{2}D_\beta D_\alpha h\end{aligned}$$

Because $D_\nu h^\nu_\alpha = D^\nu h_{\alpha\nu}$.

□

3.

$$\begin{aligned}R &= R_{\alpha\beta} g^{\alpha\beta} \implies \delta R = (\delta R_{\alpha\beta}) g^{\alpha\beta} - R_{\alpha\beta} h^{\alpha\beta} \\ &= \frac{1}{2}(g^{\alpha\beta} D_\nu D_\alpha h^\nu_\beta + g^{\beta\alpha} D_\nu D_\beta h^\nu_\alpha - g^{\alpha\beta} \square h_{\alpha\beta}) - \frac{1}{2}h^{\alpha\beta} D_\beta D_\alpha h - R_{\alpha\beta} h^{\alpha\beta} \\ &= (D_\alpha D_\beta - R_{\alpha\beta} - g_{\alpha\beta} \square) h^{\alpha\beta}\end{aligned}$$

Where we have used the fact that $g^{\alpha\beta}$ is symmetric, and we discard $\frac{1}{2}h^{\alpha\beta} D_\beta D_\alpha h$ because it has an order 2 of h .

□

Extra task: To be done...